

UNIT 4

ADAPTATIONS IN LAND ENVIRONMENTS

Learning objectives:

- Identify different biomes and describe their characteristics (desert, forest, grassland, tundra).
- Explain the basic needs of living things (water, food, air, shelter).
- Describe how plants get water, light, and nutrients to survive.
- Explain how animals get food, water, and oxygen to live.
- Understand the concept of adaptations and how they help living things survive in different environments.
- Identify adaptations of desert, forest, grassland, and arctic plants and animals.
- Compare the climates of deserts, forests, grasslands, and arctic tundra.
- Describe how seasons affect living things in different biomes.
- Recognize how animals use mimicry and camouflage for survival.
- Explain how migration helps arctic animals survive.



What are adaptations, and how do they help living things survive?

4.1 LIVING THINGS AND THEIR NEEDS

Where do living things live?

Living things live where they can meet their needs. A living thing's **environment** includes all the living and nonliving things in a place. A **biome** is a place that has certain kinds of living and nonliving things. Deserts, forests, and grasslands are types of biomes.



↑ A Buffalo eats the grass in its biome

Different biomes have different kinds of weather over time. Some biomes are cold and dry. Some are warm and wet. The soil, or ground covering, might also be different. It might be hard or soft, moist or dry.



↑ A desert biome has a hot, dry climate.



How do plants get what they need?

Most plants need the same things to live. They need water, light, food, and carbon dioxide. Carbon dioxide is found in the air. They must get all these things in their environment.



Stems carry water and food.

Roots get water and food.

Most plants have roots, stems, and leaves. Each part of a plant helps it to live. Roots get water and food from the ground. Stems carry food and water through the plant. Leaves get sunlight and carbon dioxide. Other parts, like flowers and seeds, help a plant make more plants.



Leaves get sunlight and carbon dioxide.



How do animals get what they need?

Animals also need certain things to live. They need water, food for energy, and oxygen from the air or water. They also need a place to stay safe. An animal's parts help it to live in its environment. Legs or wings help it move. Teeth or beaks help it eat.



↑ A lion uses its tongue to get water.

When animals breathe, they take in oxygen. Many animals have lungs to get oxygen from the air. Fish use gills to get oxygen from water. Other animal parts help keep them safe. They might also use their parts to help build homes and shelters. Birds use their beaks to build nests for shelter.



↑ Birds use nests for shelter.



↑ Fish use gills to breath.

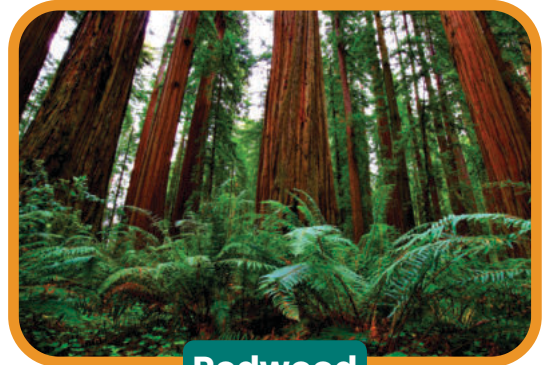


What helps living things survive in their environment?

Living things live where they can meet their needs. A redwood tree needs a cool, wet environment with rich soil. A cactus needs a warm, dry environment with sandy soil.



Cactus



Redwood

Quick Check

Circle the proper conditions for each plant in the chart below.

Plant	Temperature	Type of Soil	Amount of Moisture
1. Cactus	Warm Cool	Rich Sandy	Dry Damp
2. Redwood	Warm Cool	Rich Sandy	Dry Damp



Living things have adaptations to help them survive in an environment. An **adaptation** is a body part or a way of acting that helps a living thing survive. Plants might have long roots to reach water deep in the ground. Animals might have a certain way of finding food to eat.

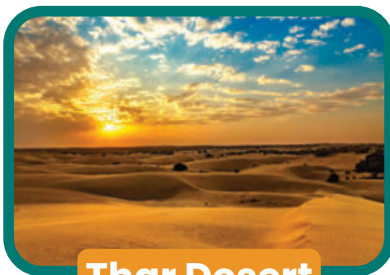


↑ A bear catches fish in a river for food.

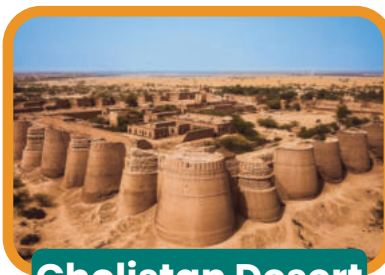
4.2 LIFE IN THE DESERT

What is a desert?

A **desert** is a biome with warm, dry weather with very little rain. In Pakistan, the main deserts are the **Thar Desert, the Cholistan Desert, and the Kharan Desert**. These deserts are very hot during the day and can become cool at night.



Thar Desert



Cholistan Desert



Kharan Desert



A desert usually gets less than 25 centimeters (10 inches) of rain in a whole year. Sometimes, it rains a lot on one day, but then there might be no rain for many months. During the day, the Sun makes the desert land and air very hot. But at night, the heat goes away quickly, and the desert becomes cool.

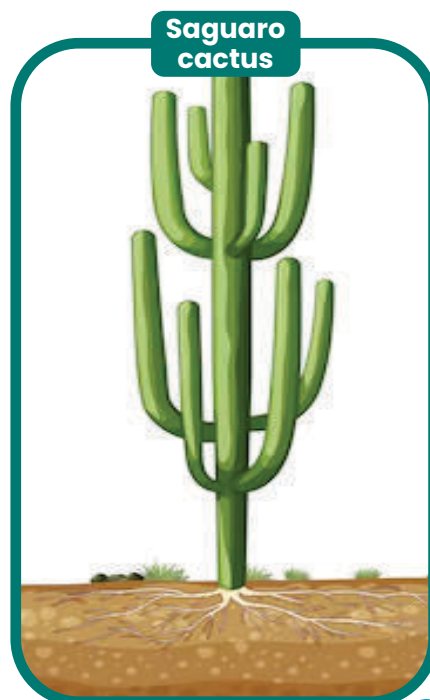
The soil in deserts is mostly sand. When it rains, the water quickly sinks into the sand. The water goes deep underground, farther than most plants' roots can reach. This makes it hard for many plants to grow in the desert. Even though deserts are very dry, some special plants and animals live there. They have adapted to survive with little water.

What adaptations help desert plants?

Desert plants have adaptations that help them get the water they need. Some plants have deep roots. These reach down into the ground for water. Other plants have shallow roots that spread out wide to quickly trap water when it rains.

Sharp spines keep animals away.

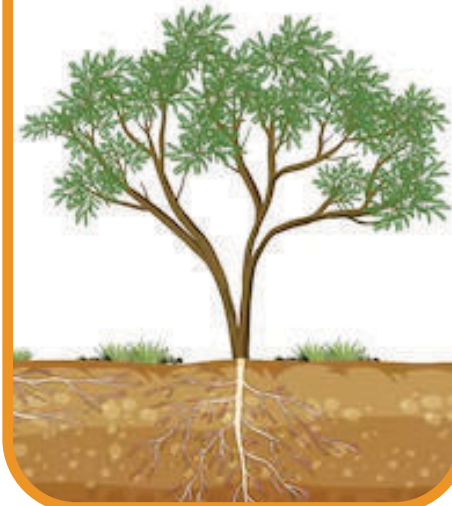
Thick, waxy stem helps store water.



Shallow roots trap water from rain.



**Mesquite
Tree**



Long roots reach deep underground.

Small leaves help save water.

Sharp thorns keep animals away.

Desert plants have other adaptations to help them survive. Thick stems store water for dry times. Waxy skin covers the plants and keeps water from escaping. Thorns and spines protect the plants by keeping animals from eating them for food and water.

For example, the aloe plant can grow well in deserts. Its waxy skin and thick leaves help store a lot of water, allowing it to survive even when there is little rain.

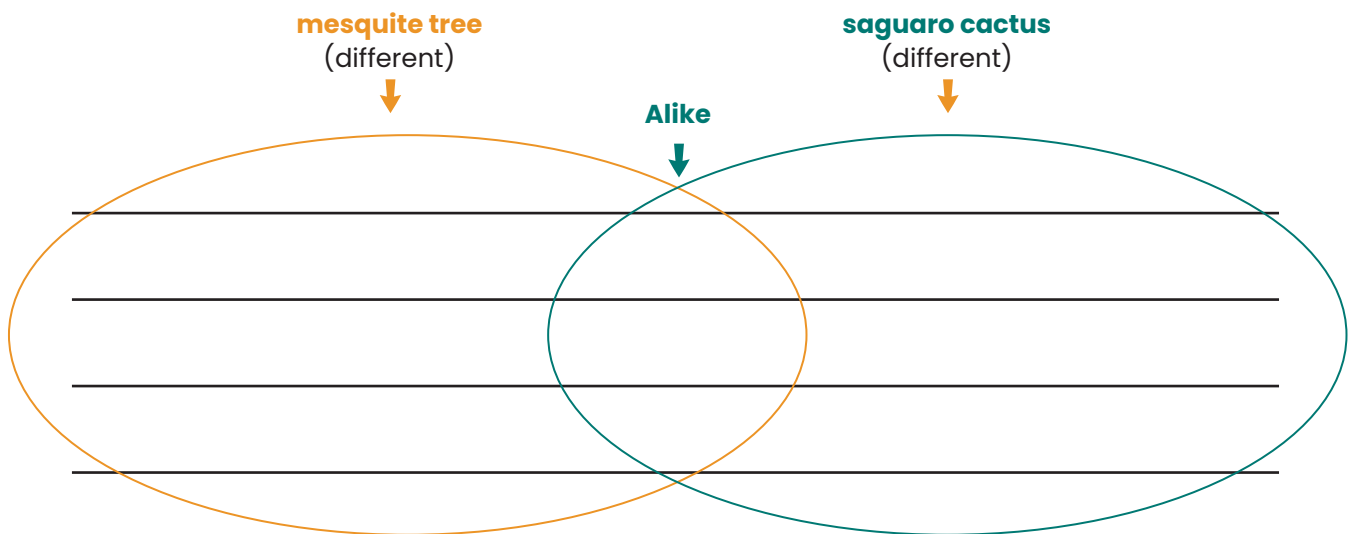


↑ An aloe's waxy skin and thick leaves are adapted to store lots of water.



Quick Check

Use the graphic organizer below to compare the adaptations of the mesquite tree and saguaro cactus.



What adaptations help animals?

Desert animals also have adaptations that help them survive in the heat. Some sleep during the hot part of the day. They come out at night when it is cool.



↑ Rattlesnakes sleep during the day.



Other animals have body parts that help them stay cool. A jackrabbit's big ears help it get rid of body heat. A thin body also helps it stay cool.



↑ This jackrabbit's big ears keep it cool.

Some desert animals can be hard to see. Their bodies blend in with their environment. They can hide easily from other animals. This adaptation helps keep them safe.



↑ This frog can blend in with its environment



4.3 LIFE IN THE GRASSLAND

What is a grassland?

A **grassland** is a biome with many grasses. It helps keep the land warm and moist. It gives animals a safe place to live and food to eat. There are two kinds of grasslands. One has mild weather and four seasons. This grassland has rich soil that helps the grasses grow. The prairies of North America are an example of this type of grassland.



↑ These grasslands in North America are called prairies.

The other type of grassland is warm all year long. These grasslands have a rainy season and a dry season. They also have trees and poor soil. During the dry season, these grasslands can dry out. They may catch fire.

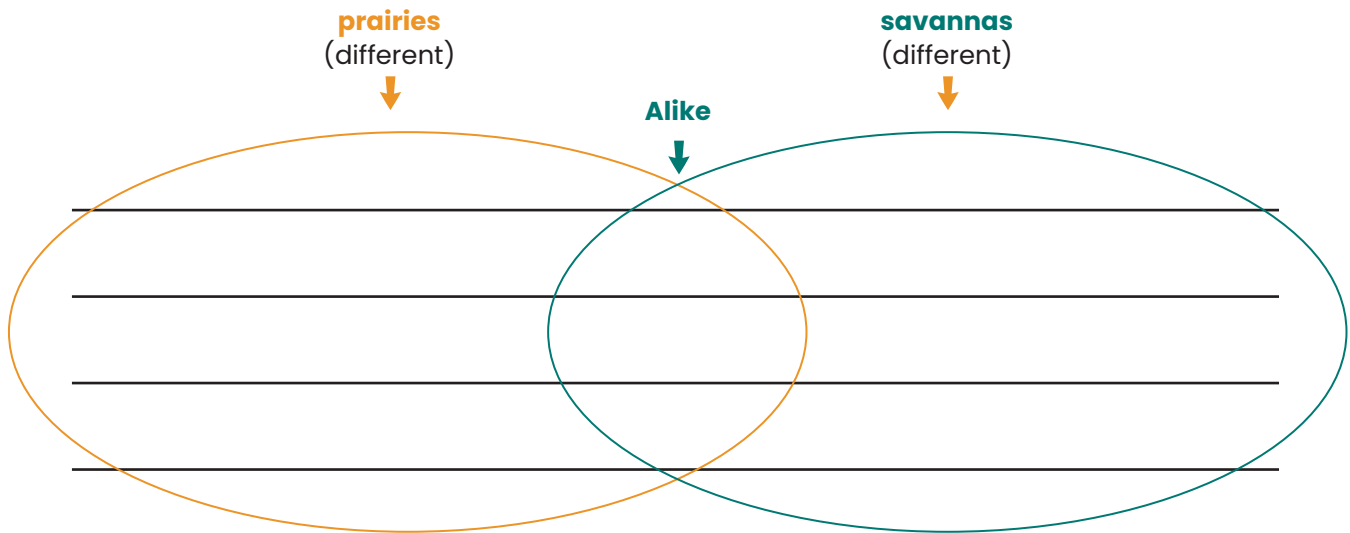


↑ These grasslands in Africa are called savannas.



Quick Check

Use the graphic organizer below to compare the prairies and the savannas.



4.4 What adaptations help grassland plants survive?

Grassland grasses are adapted to dry weather. Most grasses have deep roots. This lets them get water from deep in the ground. If fire burns the part of the plant above the ground, the roots can grow new parts. This helps the grasses stay alive in the grassland.



↑ Dry weather can lead to fires in the grassland.



Quick Check

What would a grassland look like the day after a summer fire? A month later?

Trees in the grasslands also have adaptations that help them survive. The baobab tree only grows leaves in the wet season. It has a thick trunk to help it store water in the dry season. Its bark protects it against fire.



↑ Adaptations help the baobab tree live in a grassland environment.

Quick Check

Circle the adaptation that helps the baobab tree survive in the grassland.

thin bark

thick trunk

few roots



4.5 What adaptations help animals survive in grasslands?

Animals also have adaptations that help them live in the grasslands. Some have flat teeth to help them eat the grasses. Others dig burrows, or holes, in the ground. This lets them hide from other animals. It also helps them stay cool in the heat.

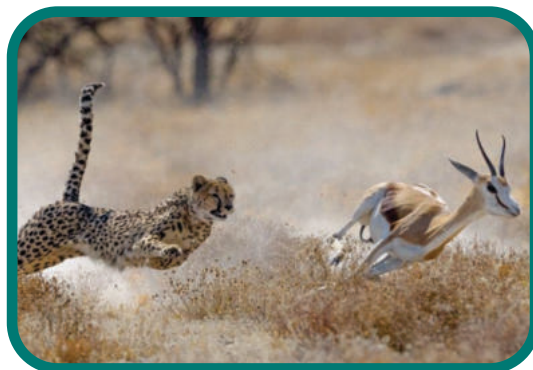


A zebra's flat teeth can chew grass.



A prairie dog can hide in a burrow.

Some grassland animals can run very fast. This helps them catch other animals for food.



↑ **A cheetah can run fast to catch food.**





Quick Check

Tell how each adaptation below helps that animal survive.

1. A prairie dog digs a burrow.

2. A zebra has flat teeth.

3. A cheetah can run very fast.

4.6 LIFE IN THE FOREST

What is a forest?

A **forest** is a biome that has many trees. There are different kinds of forests in different places around the world. In Pakistan, we have forests like the Himalayan forests in the north and mangrove forests near the coast. Forests give us fresh air, wood, and homes for animals. Some forests are always hot and wet, like tropical rainforests. Others, like temperate forests, have four seasons: spring, summer, autumn, and winter.





↑ Tropical rain forests are dense and green.

How do plants survive in forests?

Tall trees have strong roots, small plants have pointy leaves to drain water, and ground plants have big leaves to catch sunlight. In temperate forests, some trees lose leaves in winter, while others stay green.

Quick Check

Explain how plants have adaptations.

1. Plants on the ground in the rainforest get a lot of sunlight.

2. Some trees in a temperate forest have no leaves in the winter.





↑ Temperate forests change from season to season.

What adaptations help forest plants survive?

Forest plants grow in layers. Plants in each layer have adapted to get the light and water they need to live. In a tropical rain forest, tall trees have special roots that help keep them up. Some shorter plants have leaves with tips to get rid of extra water. Plants on the ground have large leaves that help them get sunlight.

Tropical Rain Forest



How do animals survive in a tropical rainforest?

Tropical rainforests have more kinds of animals than anywhere else. Their adaptations help them stay safe. Some bright, colourful animals use poison to stay safe. Their colour warns other animals to stay away. An animal that attacks might get poisoned.



↑ This frog's colours warn that it may be poisonous.

Some animals in tropical rain forests can run fast to get away from danger. Others climb high in the trees, or swim to be safe. **Mimicry** is when one thing pretends to be another thing. Some animals use mimicry to blend in with their environment. The color and body shape of a mantis make it look like a flower.



↑ This mantis looks like a flower and is hard to see. To find the mantis, look for its head.



Quick Check

Tell how each adaptation below helps forest animals.

1. Colours that stand out

2. Colours that blend in

How do animals survive in a temperate forest?

Animals in the temperate forest need to survive cold weather. Their adaptations help them. Some grow heavy coats in winter to stay warm. Some eat more or store food for the winter. Others sleep through the coldest months.



↑ This dormouse sleeps through the winter.



Animals in the temperate forest need to stay safe. Their adaptations help them survive. A skunk can spray other animals. The spray smells and stings their eyes. A porcupine has many sharp quills. These can hurt an animal that tries to attack. Most animals will leave the porcupine alone.



↑ This porcupine's quills help keep it safe



↑ This skunk can spray other animals.

4.7 LIFE IN THE ARCTIC TUNDRA

What is an arctic tundra?

The arctic tundra is a cold biome above the Arctic Circle. Winters are long, dark, and very cold. Few plants and animals can be found. In the middle of winter, the Sun never rises. In the middle of summer, the Sun never sets.



These maps show where the arctic tundra can be found.





↑ The arctic tundra has a short but beautiful summer.

Part of the tundra stays frozen all year long. Snow on top of the ground may melt in the summer. The water cannot soak into the ground. Puddles and pools of water form on top. Small plants grow on the tundra in summer. Animals come back to the tundra to eat the plants. Some build nests in the plants.

✓ *Quick Check*

Circle the correct answer.

In which direction is the Arctic tundra located?

north south east west



What adaptations help arctic plants?

Arctic plants have adaptations that help them survive. Their roots stay close to the top of the ground. This helps them get water. Some arctic plants have no roots at all. Arctic plants stay short and small. This protects them from the cold and the wind.



lichen stay short and cling to rocks.

Many plants on the tundra grow in clumps. This protects them from the wind and cold. The dark colors of arctic flowers also help the plants. They attract animals that can help spread seeds. This lets new plants grow.



↑ The dark flowers of the dwarf fireweed help it attract animals.



Quick Check

Fill in the blank.

Most arctic flowers have a _____ colour.

How do animals survive in the arctic?

Arctic animals have adaptations that help them stay warm. Some have small legs and arms to keep them from losing heat. Some have heavy fur that protects them from the cold. A thick layer of fat keeps their bodies warm. Polar bears have large bodies to keep heat, and animals like lynxes have wide feet to walk on snow.



↑ This polar bear has adaptations that keep it warm.

Quick Check

Why would a polar bear in the tundra have a larger body than a black bear in the forest?



Other adaptations help arctic animals move on snow and ice. Some have wide feet. This helps them stay on top of the snow. Sharp claws keep them from slipping on ice.

What are some other arctic animal adaptations?

Many arctic animals **migrate** each year. They move from one place to another. Caribou, tundra swans, and Canadian geese are examples. These animals cannot survive the bitter cold winters. So they migrate south where it is warmer, and where they can find food. The arctic tundra will get warmer in spring. Plants will grow again. Animals that migrated south will return.



↑ Animals such as the caribou migrate each year.



Some arctic animals change colour with the seasons. Animals, like the arctic fox, change colour to white in winter to blend with the snow.



↑ Arctic fox in summer.



↑ Arctic fox in winter



Use the words from the word box to fill in the blanks.

environment

grassland

forest

desert

tundra

adaptation

mimicry

biome

migration

1. A(n) _____ has certain kinds of living and non living things.
2. Animals that move from one place to another, is called _____.
3. A(n) _____ includes all the living and nonliving things in a place.
4. A(n) _____ has many trees.
5. _____ lets one animal look like another animal or plant.
6. A(n) _____ is a biome with many grasses.
7. A thick coat is a(n) _____ that helps an animal survive cold weather.
8. The arctic _____ is a cold biome in the far north.
9. A(n) _____ has hot, dry weather.



Test Your Understanding

Q1. What is a biome?

Ans: _____

Q2. Define desert and grassland.

Ans: _____

Q3. Name the things plants need to live.

Ans: _____

Q4. How do roots help a plant survive?

Ans: _____

Q5. What do animals need to live?

Ans: _____





Q6. How do desert plants store water?

Ans: _____

Q7. Why do some desert animals sleep during the day?

Ans: _____

Q8. What helps grassland animals stay safe from danger?

Ans: _____

Q9. Why do arctic animals have thick fur?

Ans: _____

Q10. How does the white fur of an arctic fox help it survive in winter?

Ans: _____



Fill in the blanks.

1. A _____ is a large area with certain kinds of plants and animals.
2. Plants use their _____ to take in water from the soil.
3. Animals need _____ from the air or water to breathe.
4. Desert plants have _____ stems to store water.
5. A cactus grows in _____ soil because it holds little water.
6. _____ helps animals blend in with their surroundings.
7. Grassland animals dig _____ to stay cool and hide.
8. Forests in Pakistan include Himalayan forests and _____ forests.
9. Arctic animals have a thick layer of _____ to keep warm.
10. The _____ Desert is the largest desert in Pakistan.



Tick (✓) for True and cross (x) for False statements.

1. Deserts get a lot of rain throughout the year. ()
2. Grasslands have many tall trees. ()
3. Plants need carbon dioxide to make food. ()
4. Animals use mimicry to attract predators. ()
5. Arctic tundra is warm all year round. ()
6. Forests provide homes for many animals. ()
7. Some animals store food to survive the winter. ()
8. Desert animals are active during the hottest
part of the day. ()
9. A cheetah can run fast to catch its food. ()
10. The arctic fox changes its colour in winter to
blend in with snow. ()



Multiple Choice Question.

1. Why do arctic animals have small legs and arms?

- a) To help them swim faster
- b) To reduce heat loss
- c) To make them look smaller

2. What helps desert plants store water?

- a) Thin stems
- b) Waxy skin
- c) Bright flowers

3. Which adaptation helps animals stay on top of the snow?

- a) Sharp claws
- b) Wide feet
- c) Long tails

4. What is a major reason of animals migration?

- a) To escape predators
- b) To find warmer places and food
- c) To grow new fur



5.How do dark coloured arctic flowers benefit?

- a) They attract insects
- b) They repel insects
- c) They grow faster in the shade

6.Which biome is characterized by hot, dry weather?

- a)Tundra
- b) Forest
- c) Desert

7.What helps rainforest plants get sunlight?

- a) Growing tall to reach light
- b) Having thick bark
- c) Growing underground

8.How do animals in rainforests escape danger?

- a) By hibernating
- b) By climbing trees or swimming
- c) By growing thick fur





Biome in a Box



ACTIVITY

Create a small diorama representing one of the biomes (tundra, desert, forest, or grassland). Use materials like cotton for snow in the tundra or sand for the desert. Add small animal and plant figures to show their adaptations. Label the adaptations and explain how they help living things survive in that environment.

